FSR and IMU sensors-based human gait phase detection and its correlation with EMG signal for different terrain walk

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Abstract

Purpose – The purpose of this paper is to present gait analysis for five different terrains: level ground, ramp ascent, ramp descent, stair ascent and stair descent.

Design/methodology/approach – Gait analysis has been carried out using a combination of the following sensors: force-sensitive resistor (FSR) sensors fabricated in foot insole to sense foot pressure, a gyroscopic sensor to detect the angular velocity of the shank and MyoWare electromyographic muscle sensors to detect muscle's activities. All these sensors were integrated around the Arduino nano controller board for signal acquisition and conditioning purposes. In the present scheme, the muscle activities were obtained from the tibialis anterior and medial gastrocnemius muscles using electromyography (EMG) electrodes, and the acquired EMG signals were correlated with the simultaneously attained signals from the FSR and gyroscope sensors. The nRF24L01+ transceivers were used to transfer the acquired data wirelessly to the computer for further analysis. For the acquisition of sensor data, a Python-based graphical user interface has been designed to analyze and display the processed data. In the present paper, the authors got motivated to design and develop a reliable real-time gait phase detection technique that can be used later in designing a control scheme for the powered ankle-foot prosthesis.

Findings – The effectiveness of the gait phase detection was obtained in an open environment. Both off-line and real-time gait events and gait phase detections were accomplished for the FSR and gyroscopic sensors. Both sensors showed their usefulness for detecting the gait events in real-time, i.e. within 10 ms. The heuristic rules and a zero-crossing based-algorithm for the shank angular rate correctly identified all the gait events for the locomotion in all five terrains.

Practical implications – This study leads to an understanding of human gait analysis for different types of terrains. A real-time standalone system has been designed and realized, which may find application in the design and development of ankle-foot prosthesis having real-time control feature for the above five terrains.

Originality/value – The noise-free data from three sensors were collected in the same time frame from both legs using a wireless sensor network between two transmitters and a single receiver. Unlike the data collection using a treadmill in a laboratory environment, this setup is useful for gait analysis in an open environment for different terrains.

Keywords Electromyography, Wireless sensor network, Force-sensitive resistors, Gait phase detection, Inertial measurement units

Paper type Research paper

1. Introduction

Gait analysis is the systematic investigation of human locomotion (Tao *et al.*, 2012). The term human gait defines the way of human walking, and it comprises of successive gait cycles. A gait cycle is well-defined as the duration from one heel strike to the subsequent heel strike on the same limb. The gait cycle divides into two main phases, the stance and swing (SW) phases (Winter, 2009). The stance phase represents the time for which a foot is on the ground, whereas the SW represents the total time during which the foot is in the air. Figure 1 shows the gait cycle for the right reference lower limb. The gait cycle further splits into eight sub-phases, i.e. initial contact (IC), loading response (LR), midstance (MSt), terminal stance (TSt), pre-swing (PSw), initial swing (ISw), mid-swing (MS)

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Sensor Review
41/3 (2021) 235–245
© Emerald Publishing Limited
[ISSN 0260-2288]
[DOI 10.1108/SR-10-2020-0249]

This research is partially supported by Cisco thingQbator, MCIIE IIT (BHU), Cohort-3/IIT21. The authors would like to thanks Milind Prajapat, Ujjwal Sagar and Kartik Garg for their support during the experimental work.

Conflicts of interest: The authors declare that there are no conflicts of interest.

Ethics approval: This study is approved by the ethical committee of the Institute of Medical Sciences, BHU, Varanasi.

Received 30 October 2020
Revised 26 December 2020
Accepted 7 February 2021

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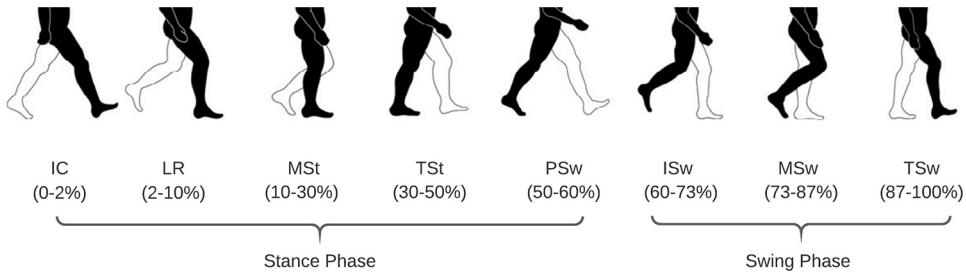
Figure 1 Gait cycle (Stride)



and terminal swing (Winter, 2009; Prakash *et al.*, 2018). Whittle (2014) used heel-contact, foot-flat, mid-stance, heel-off, toe-off (TO) and MS events to divide a gait cycle into convenient periods.

Available gait approaches can be categorized into vision-based, sensor-based and hybrid approaches (Prakash *et al.*, 2018). Non-wearable and wearable sensors-based devices are available for human gait analysis. Non-wearable sensors-based devices are most suitable for gait analysis in the laboratory environment. It includes an image processing-based system and a floor sensors-based system to capture gait. However, it is possible to capture human gait data outside the laboratory during everyday activities using wearable sensors-based devices. Wearable sensor-based gait recognition resulted in widespread utilization of clinical diagnosis, rehabilitation and sports. Tao *et al.* (2012) have presented the available wearable sensors and ambulatory gait analysis methods. The wearable sensors-based devices use sensors placed on the subject's feet, knees and thighs. Sensors such as accelerometers, gyroscopes, force sensors, electro-goniometers and electromyography (EMG) are commonly used wearable sensors to characterize human gait (Muro-De-La-Herran *et al.*, 2014; Anwary *et al.*, 2018). Magnetic system-based approach, electro goniometer, force myography (FMG) and imaging modalities (magnetic resonance imaging, computed tomography and ultrasound medical imaging) are other techniques for human gait analysis. A combination of two or more approaches mentioned above is termed a hybrid approach.

Wearable sensor-based gait analysis techniques generally fall into gait kinematics, kinetics and EMG. The kinematics of the human gait defines the activities of the major joints and components in the lower limbs. Gait kinetics highlights the study of forces and moments that result in the human segment's movement. The human gait includes changes in the angles of each joint in the lower limbs. These angles change with changes in the bioelectrical energy of skeletal muscles. Therefore, the EMG signal generated during human gait relates to the posture angle of each joint during human gait. In the present study, authors used force-sensitive resistors (FSRs) and inertial measurement unit (IMU) sensors simultaneously for gait phase detection. Also, the EMG signals are acquired in the same time frame from two leg muscles, where the waveform patterns are explained in terms of flexion and extension of the ankle during human locomotion on different terrain. FSRs sensors of different kinds and sizes were used by many authors to be used in smart insoles either to measure foot pressure during walking or as footswitches (Aminian *et al.*, 2002; Pappas *et al.*, 2004; Malvade *et al.*, 2017). Godiyal *et al.* (2018) have acquired FMG signals from the thigh to determine the heel-strike (HS) and TO events during overground and ramp walking. The FlexiForce is a kind of standard tiny thin FSR whose resistance varies with pressure. The authors used 4-FlexiForce A401 sensors placed on each foot insole for gait phase detection in this study.

Another mostly used sensor for the gait analysis is the IMU sensor that provides the measurement of acceleration and gyroscopic signals from the body during locomotion (Sabatini *et al.*, 2005; Joshi *et al.*, 2014; Muller *et al.*, 2015; Gouwanda and Gopalai, 2015; Gouwanda *et al.*, 2016; Maqbool *et al.*, 2016). Sabatini *et al.* (2005) have devised a gait phase segmentation technique by integrating IMU to determine off-line temporal gait parameters using MATLAB. Gouwanda *et al.* (2015) have acquired gyroscope data from InvenSense MPU 6000 to identify HS and TO gait events and determine the temporal gait parameters. The peaks greater than zero correspond to MS, and troughs less than zero correspond to HS and TO in the curve characteristics. The authors used algorithms based on heuristics and zero-crossing methods for gait event detection in the present paper. The first derivative of the angular rate was used for monitoring zero crossings, which correspond to a potential MS, HS and TO. When a zero crossing or a sign change was detected, the corresponding angular rate was recorded as a potential point. Finally, heuristics were used to evaluate these potential points to determine the related gait events.

Technological improvements have led to numerous wearable devices for the physical support and restoration of human locomotion (Tucker *et al.*, 2015). Orthosis describes a device that is used for supporting a person with limbs impairment. Blaya and Herr (2004) have presented an active ankle-foot orthosis where the orthotic joint impedance is controlled during stride for analyzing drop-foot gait. Another boon to the amputees is prosthetic devices that replace a missing part of the body, and it plays an essential role in rehabilitation (Rietman *et al.*, 2002). There is excellent scope for improving lower-limb prostheses that require collaborative research from multidisciplinary subjects such as biomedical, mechanics, mechanical, electrical and mechatronics (Kumar *et al.*, 2017). The ankle joint receives the body's whole weight; hence, the ankle's biomechanical analysis during gait on the ground level, slopes and stairs plays a significant role while designing ankle-foot prosthesis. This paper presented the off-line and the real-

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time experimental results of gait analysis using FSR, IMU and EMG sensors for different walking terrains in an open environment. This study is useful later for the control of an ankle-foot prosthesis.

2. Materials and method

2.1 Sensor module

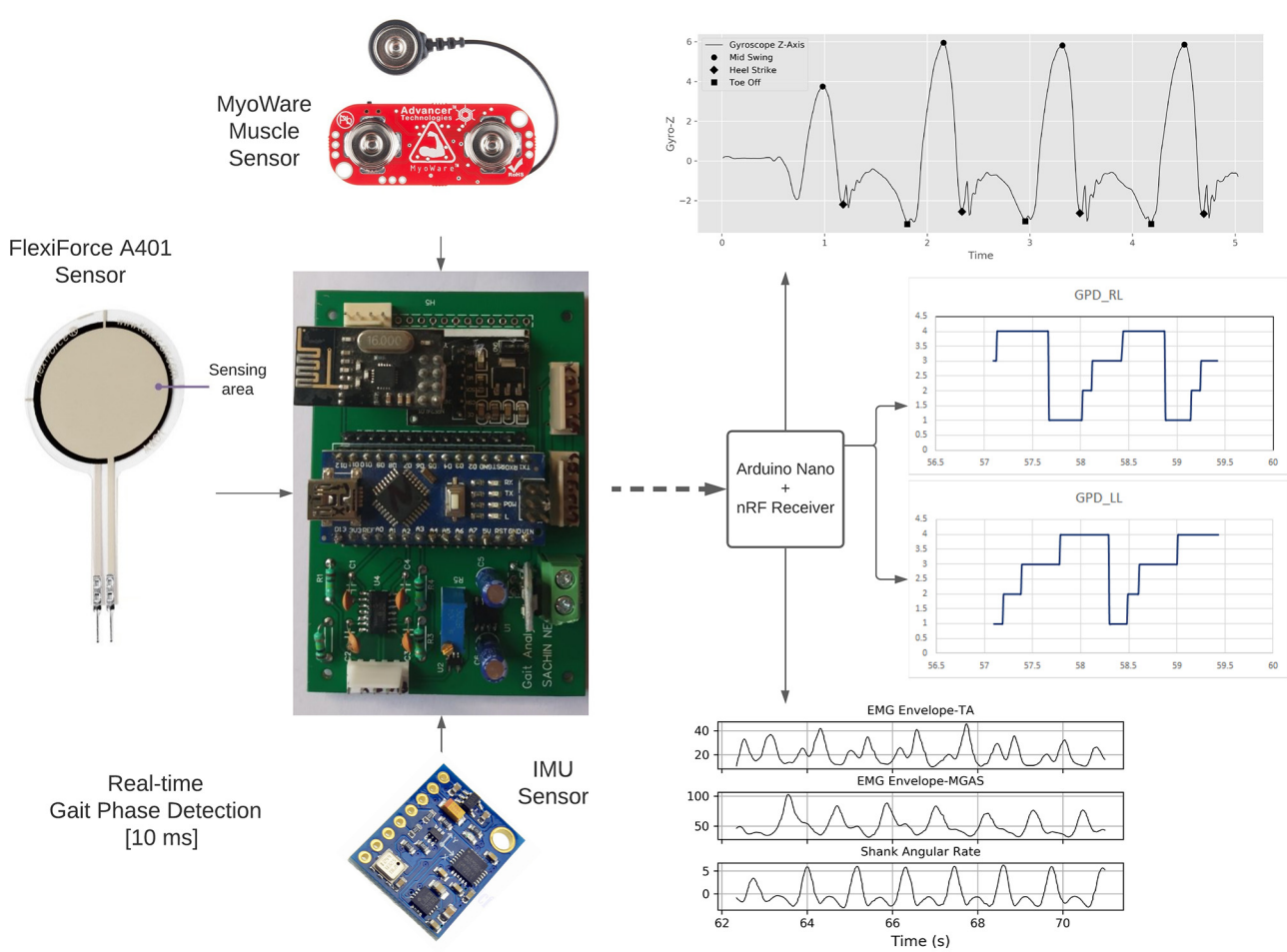
Figure 2 illustrates the wireless setup designed for acquiring the signals from the sensors. The sensor module used in the present study for gait analysis consists of FSR, IMU and EMG sensors. FlexiForce A401 [Tekscan Inc.], a standard form of FSR sensors, was used in the present study as footswitch to determine gait phase parameters, and it also serves to detect the foot pressure after proper calibration. A GY-87, 10-DOF IMU sensor board was featured with a 3-axis accelerometer, 3-axis gyroscope, 3-axis magnetometer and barometric pressure sensor. EMG signals were acquired from the tibialis anterior (TA) and medial gastrocnemius (MGAS) muscles of the leg. The surface EMG (sEMG) signal acquired from TA muscles represents dorsiflexion, whereas the signal from MGAS represents the foot's plantar-flexion. Arduino nano controller board read the FSR sensors data from the foot insole, IMU accelerometer/gyroscopic signals from the shank, and the sEMG signals from TA and MGAS muscles. The nRF24L01+ wireless modules were used to transmit data between Arduino nano and the computer.

2.2 Signal conditioning

FlexiForce A401 sensor is a piezoresistive force sensor with a 25.4 mm sensing area. The primary circuit for the detection of the FlexiForce sensor signal is shown in Figure 3. To generate stable negative V_{REF}, an ICL7660 charge-pump IC was used, followed by a voltage divider and LM321 OpAmp IC. By changing the V_{REF} voltage, one can modify the dynamic range of this force sensor. All the FlexiForce sensors were calibrated for different weights before installing them on the shoe insole. Figure 4 shows the weight versus resistance curve and the weight versus voltage calibration curve for four-FlexiForce sensors. The sensor's electrical output is related to the actual weight applied to the sensing area by doing the calibration.

In the GY-87 board, the MPU6050 device integrates a three-axis gyroscope and a three-axis accelerometer on the same silicon chip collectively with an onboard Digital Motion Processor capable of processing complicated nine-axis Motion Fusion algorithms. The sEMG signals were acquired using wet surface electrodes, where raw signals were fed to a low power instrumentation amplifier that offers excellent

Figure 2 Wireless system for gait analysis



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accuracy with very low offset voltage, drift and high common-mode rejection. After that, the signal is passed through a 20–500 Hz bandpass filter to remove the motion artifacts, external noises of low frequencies and noises occurring at high frequencies (De Luca, 2002; Hermens *et al.*, 2000). An envelope detector containing a precision rectifier circuit and a low-pass resistor–capacitor circuit was used to yield a linear envelope of amplified and filtered EMG signal (Prakash *et al.*, 2019).

2.3 Gait phase detection

2.3.1 Gait phase detection using force-sensitive resistor sensors

Stance and SWs can be detected easily using two FSR sensors. One FSR was attached below the heel and another one below the toe position. Figure 5 shows the flow chart for the detection of these gait phases. The method consisted of a threshold depending on the weight of the subject and a set of heuristics. The code was implemented in the Python language. Here, FF_{heel} and FF_{toe} denote the output of FSR sensors attached at heel and toe positions, respectively. The algorithm first detects the HS event, whose timing and magnitude are represented by hs_ptt and hs_ptm. Once the HS point is detected, the algorithm search for the TO point, which is represented by to_ptt and to_ptm. By assuming the HS occurred before the TO, the swing and stance phases were obtained using the following equations:

$$t_{gait}[j] = hs_ptt\,[j+1,\ 0] - hs_ptt\,[j,\ 0], \tag{1}$$

$$t_{swing}[j] = hs_ptt\,[j+1,\ 0] - to_ptt\,[j, 0], \tag{2}$$

$$t_{stance}[j] = t_{gait}[j] - t_{swing}[j]. \tag{3}$$

Two sensors can detect only stance and SWs. Au *et al.* (2008) used four phases in a gait cycle: controlled plantar-flexion (CP), controlled dorsiflexion (CD), powered plantar-flexion (PP) and SW. The ankle undergoes a CP intending to lower the foot to the ground. It starts with HS and ends with a flat foot. The ankle begins to dorsiflex (CD) from the toe-strike, and it stays till maximum dorsiflexion. Later in the stance phase, PP begins at maximum dorsiflexion, during which the ankle releases the stored elastic energy for the forward propulsion of the body, and it terminates at TO. SW starts at TO and ends at HS for a level ground walk.

Figure 3 Signal conditioning for FlexiForce piezoresistive force sensor

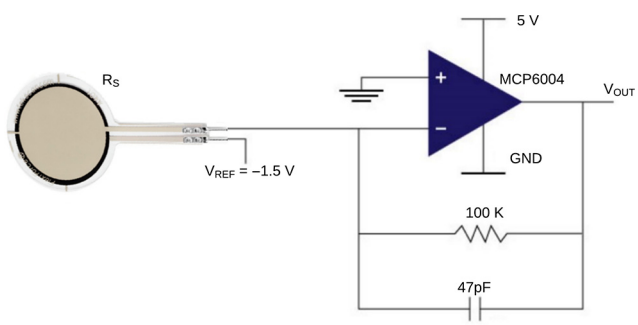

Figure 4 (a)Weight versus resistance, and (b)weight versus voltage curves for FlexiForce sensors

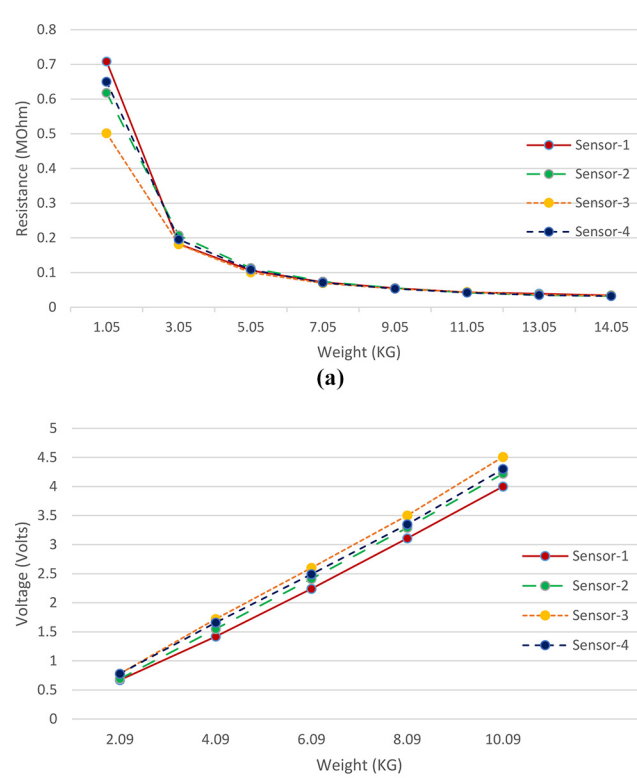

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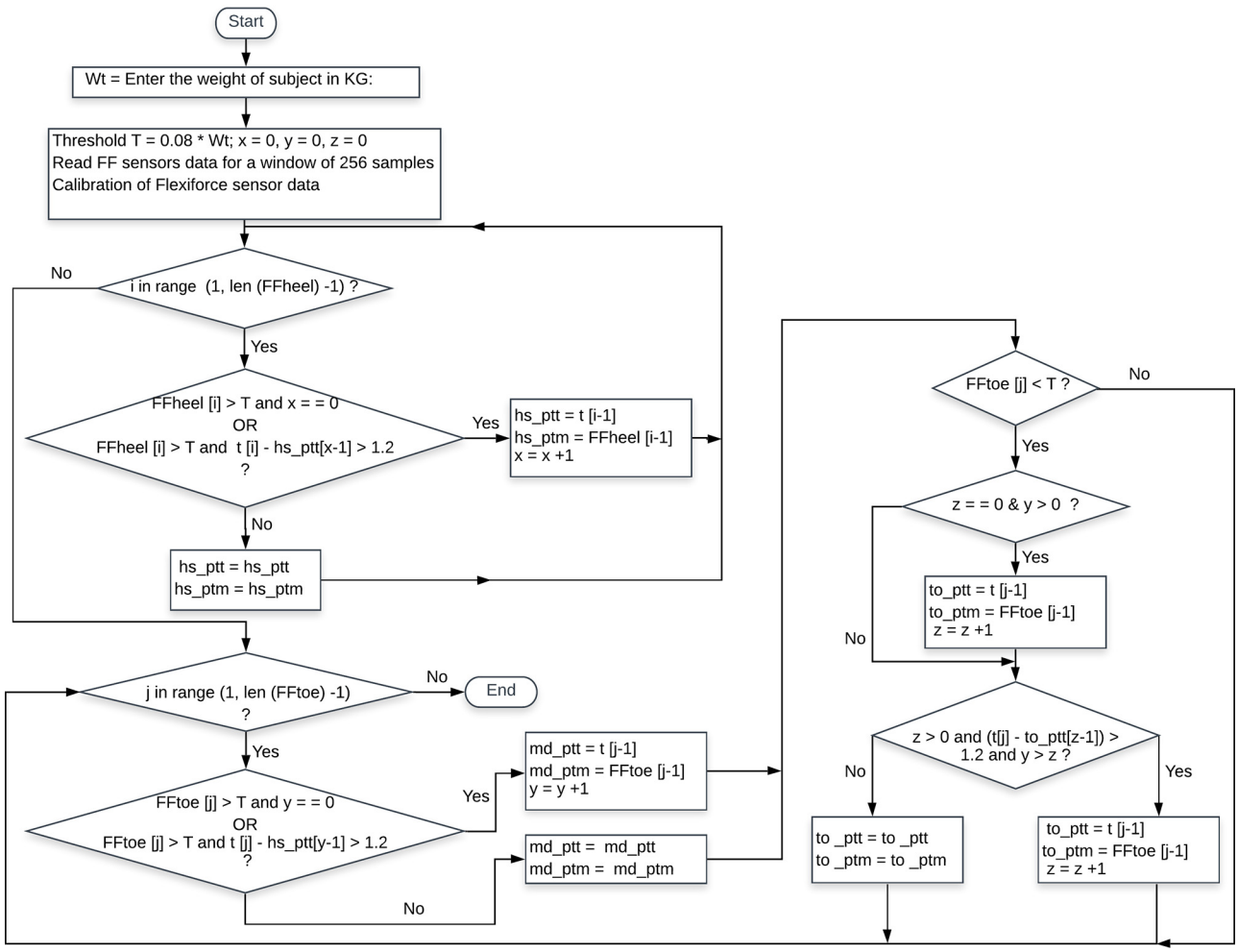
Figure 5 Flow chart of the gait phase detection algorithm using the FSR sensor



For these four phases, a real-time algorithm was realized to run on the Arduino nano board. Here four FSR sensors were used beneath the heel, toe, metatarsal-1 (MT-1) and metatarsal-5 (MT-5) region of both feet. The algorithm is as follows:

- Initialize the Arduino ports A0 to A3 as input ports.
- Read FSR data from the heel, toe, MT-1 and MT-5 positions.
- Apply the filter equation $y(n+1) = 0.9391 \times y(n) + 0.0609 \times u(n)$, that corresponds to low-pass digital filter having the cut-off frequency of 1 Hz and sampling rate 100 samples/s.
- Assign each FSR sensors either 1 or 0 depending on the threshold as below:

if (FSR_n >= threshold)

$$FSR_n = 1$$

else

$$FSR_n = 0;$$

- Write the set of rules for gait phase detection (GPD) in real-time, where the value of GPD is 1–4, corresponding to the above four phases.

if ((Heel = 1) and (MT1 = 0) and (MT5 = 0) and (Toe = 0))

$$GPD = 1; //CP$$

else if ((Heel = 1) and (MT5 = 1))

$$GPD = 2; //CD$$

else if ((Heel = 0) and ((MT1 = 1) or (MT5 = 1) or (Toe = 1)))

$$GPD = 3; //PP$$

else if ((Heel = 0) and (MT1 = 0) and (MT5 = 0) and (Toe = 0))

$$GPD = 4; //SW$$

Go to step (ii).

2.3.2 Gait phase detection using inertial measurement unit sensors

A single IMU sensor was also placed at the shank to detect stance and SWs by acquiring the lower limb's angular velocity. The sensor was placed on the right leg's lateral side (medially to left leg) by pointing the z-axis outwards perpendicular to the sensor module's surface. The angular

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velocity waveform shows one maxima and two minima; these correspond to MS, HS and TO, respectively. Figure 6 shows a detailed flow chart to detect gait events.

The algorithm is as follows:

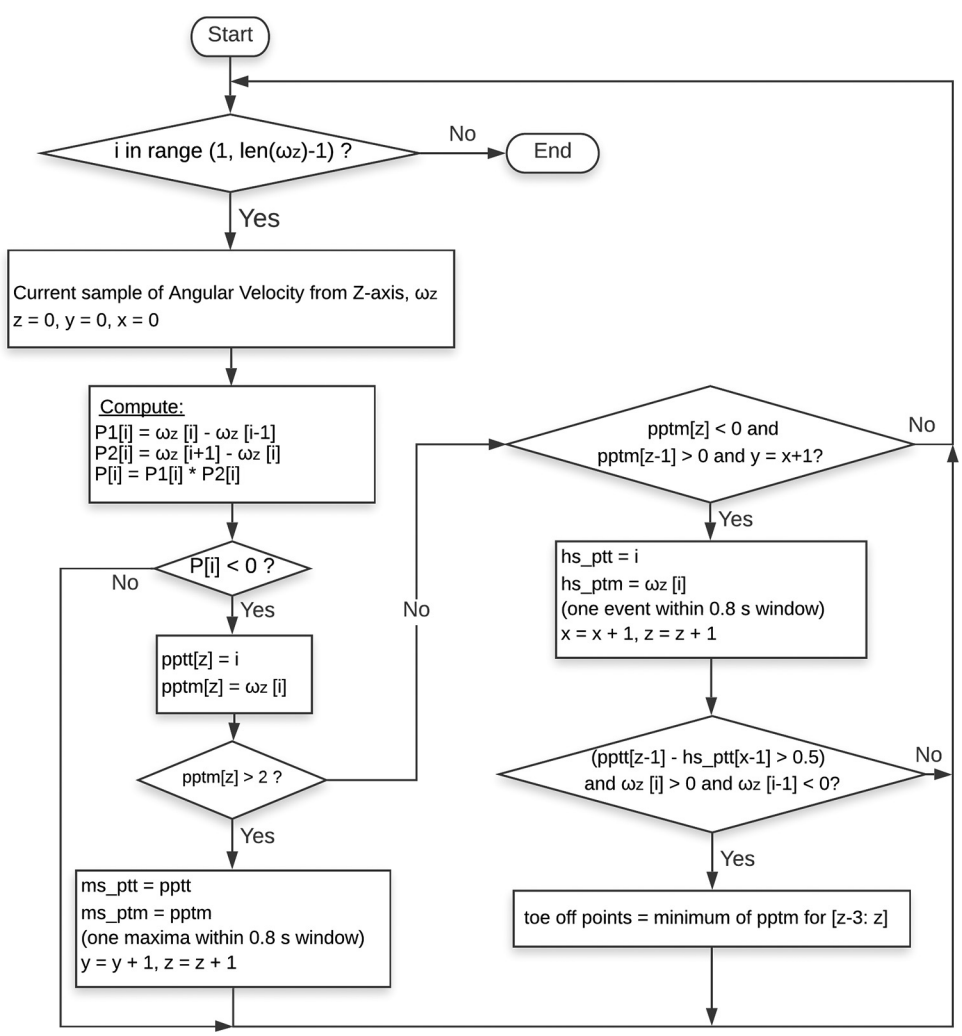
- Read the shank angular rate (ω_z).
- Obtain the potential points where the slope changes its sign in the waveform.
- Identify the potential points where the signal's amplitude is greater than a threshold (2 rad/s). Only one maximum point is allowed between the threshold interval (0.8 s), and mark these points as MS points.
- Search for two minima points after the detection of an MS point. Identify the first minima using zero crossings that are termed as HS points. Only one HS point is allowed between the threshold interval (0.8 s).
- Next, identify the second minima after that ω_z crosses zero points.
- Obtain the stance and SWs using equations (1)–(3).
- Go to step (i).

2.4 Wireless sensor network

Wireless sensor networks (WSNs) are a self-organized distributed systems built through communication with many sensor nodes in the sensory area. It incorporates recent sensors and network hi-tech with embedded computing and distributed information processing technology. Arduino is a microcontroller with increased fame in recent years, and it is an open-source platform to build an electronic prototype with high flexibility, low price and ease to use. Unlike wired network designs, wireless network designs create more flexibility during the acquisition of sensor data. For human gait phase detection in an open environment, wireless sensor technology gives hassle-free data recording and movement freedom as compared to wired connections. So, the authors have implemented this and described the same for real-time gait phase detection. The core of the WSN consists of the base node (coordinator) and end devices.

Bluetooth, ZigBee and nRF24L01+ are commonly used wireless protocols for WSNs. The nRF24L01+ transceiver module is intended to function in a 2.4 GHz ISM frequency band and listen to six other devices. The data transfer rate

Figure 6 Flow chart of the gait phase detection algorithm using the IMU sensor



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can be one of 250kbps, 1Mbps and 2Mbps. A star topology was used in the present study, where a sensor module was attached to each leg. Each module consists of four-FSRs, an IMU sensor, two-EMG sensors, an Arduino nano and an nRF24L01+ transceiver. Both devices act as transmitters. A set of an Arduino nano and nRF24L01+ transceiver acts as the receiver, connected to the computer by using a serial port. For off-line gait analysis, the Tera Term software was used to save the serial data from the receiver module in text format on the computer, which can then be extracted for later analysis in Python. For real-time gait phase detection, Arduino-based code was used for FSR sensor data; however, the algorithm was coded in Python for IMU data.

3. Results and discussion

Data were recorded from six healthy subjects (five male and one female), where the subject's mean (±STD) age, height and weight were 27 (±9.2) years, 166 (±8.8) cm and 70.5 (±12) kg, respectively. The length of the level ground, ramp and stair terrains used in this study were 34, 8 and 9 feet, respectively. The height of each stair was 20 cm, whereas the ramp slope was 8 degrees. A subject-specific foot insole was designed to record the FSR sensors data, where 30-trials data of a subject were mentioned for the analysis. Figure 7 shows the two FSR sensors' responses placed on each leg's heel and toe sites. The results show that all the events, i.e. the MS, HS and TO, are correctly identified. Using equations (1)–(3), the swing and stance phases were obtained, and these are mentioned in Table 1. For the level ground walk, the SW was detected 41% of the gait cycle, and the stance phase was found 59% of the gait cycle. Later by placing the four FSR sensors at the heel, MT-1, MT-5, and toe positions on the insole, a real-time gait phase detection technique was implemented on the Arduino code itself. The results are shown in Figure 8 for the right leg, where GPD values 1, 2, 3 and 4 correspond to CP, CD, PP and SW phases, respectively. The gait phases were identified 90% correctly for the right leg for a 30-trials level ground walk. The algorithm took approximately 10 ms time using Arduino nano for the detection of gait phases. Gait phase detection within 10 ms is well-thought-out real-time detection for the control of the prosthetic leg.

This paper also enlightens the utilization of IMU sensors to record stance and SWs. Table 2 shows that for the level ground walk, the SW detected was 40% of the gait cycle,

Table 1 Gait parameters using two-FSR sensors

Leg	t_gait (s)	t_swing (s)	t_stance (s)
Right	1.298 (±0.046)	0.535 (±0.032)	0.763 (±0.029)
Left	1.290 (±0.049)	0.532 (±0.032)	0.758 (±0.029)

Figure 7 Gait phase detection using 2-FSR sensors (a) Right leg, (b) Left leg

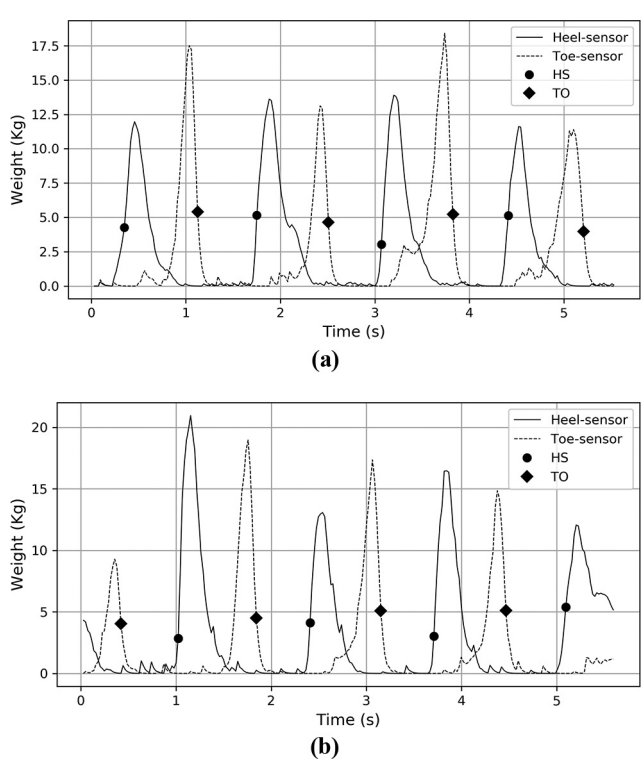

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Figure 8 Real-time gait phase detection using 4-FSR sensors

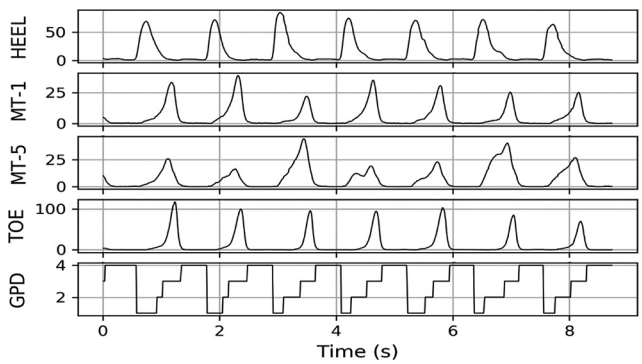

Table 2 Gait parameters using IMU sensors

Leg	t_gait (s)	t_swing (s)	t_stance (s)
Right	1.284 (±0.035)	0.526 (±0.025)	0.758 (±0.032)
Left	1.286 (±0.029)	0.504 (±0.018)	0.782 (±0.027)

and the stance phase was found 60% of the gait cycle. In Negi *et al.* (2020a), the authors used acceleration signal-based identification of gait events, where the hybrid sensors (EMG + 3-axis acceleration) were placed at TA and MGAS muscles to record EMG as well as acceleration. A single IMU sensor was placed at the lateral side (for right leg) of the shank in the present work and measured its angular rate. Figure 9 shows the shank's angular rate for walking on five different terrains. The first subplots represent the raw gyroscope signal from the z-axis of the IMU sensor, the second subplots are the low-pass filtered signal at 3 Hz cut-off, and the past subplots show the gait events marked on the signal. All the events were detected correctly, and these findings are in agreement with (Gouwanda *et al.*, 2016).

IMU sensor was also checked for real-time gait phase detection. The total time required to detect any gait event/phase is the sum of code processing time in Python and the time taken to acquire the serial monitor data. Figure 10 shows the gyroscope waveform, where the marked events were obtained within an average of 10 ms. For a 30-trials data set from the right leg, all the events were identified correctly for a level ground walk, which shows that using a single IMU attached to the shank is more favorable to collect gait data in real-time. The authors also obtained all the events correctly identified for six subjects for all five terrains. Our findings are much better than 100 ms reported by the authors (Gouwanda *et al.*, 2016). The IMUs are advantageous over FSRs in terms of cost as a single IMU sensor can detect gait- phases in real-time, and it does not require wearing a subject-specific foot insole.

The EMG sensors' response was checked simultaneously to correlate the EMG signal with the gait signal analyzed previously. Figure 11 shows the EMG signal envelope acquired from the TA and MGAS muscles for five terrains. The advantage of taking the EMG signal envelope is that one can easily acquire the EMG signal using the Arduino nano controller board, whose sampling is not as good as other standard high-cost data acquisition cards.

In Figure 11(a), TA muscle shows its action commencing at TO and stays all over the SW; but the scale is reduced during MS. TA muscle is still active at HS, and it is active until the foot completely touches the ground. The MGAS muscle is active between the initial and terminal stances. Figure 11(b) and (c) show that for descending on the ramp surfaces, TA muscle is more dominant, whereas MGAS muscle is more active during TO for ramp ascent. These waveforms are similar to the results (Pickle *et al.*, 2016), where the functional roles of different muscles, including TA and MGAS, are explained for different positive and negative slopes. For walking down on the stairs, the MGAS muscle is in peak action after the SW to make proper foot landing [Figure 11(d)]. The TA muscle is mostly active during the ISW, and it is active all through the stance phase. These findings are in agreement with (Benedetti *et al.*, 2012). Figure 11(e) represents that for stair ascent walk, the TA muscle is energetic during the SW, and it makes available proper foot clearance from the ground. MGAS muscle is active for most of the stance phase; however, there is no activity detected during the initial stance, as the heel is not entirely placed on the stairs during SA. All the results agree with another study done by the authors (Negi *et al.*, 2020a) using standard Delsys Trigno wireless EMG sensors.

For the intent-based recognition of human locomotion, the EMG signal is beneficial. It is also useful for the control of orthotic and prosthetic devices. The present study will be useful in our further work by including both the EMG sensor data acquired from the below-knee amputee's residual muscles and the FSR and IMU data collected from the artificial leg to control a power-driven ankle prosthesis. These real-time gait phase detection techniques and the real-time classifier such as support vector machine (Negi *et al.*, 2020b) find the best scope to control the prosthetic leg on different terrains.

4. Conclusions

The human gait analysis was performed using wearable and wireless sensors for five different terrains. The analysis was done with three wearable sensors, namely FSR, IMU and EMG sensors. Both off-line and real-time gait events and gait phase detections were accomplished for FSR and IMU sensors. FSR and IMU sensors showed their usefulness for detecting the gait events in real-time, i.e. within 10 milliseconds. The IMUs are advantageous over FSRs in terms of cost as a single IMU sensor can detect gait-phases in real-time, and it does not require wearing a subject-specific foot insole. Heuristic rules and a zero-crossing based-algorithm for the shank angular rate correctly identified all the gait events for the locomotion in all five terrains. This paper explains the usefulness of the real-time gait phase-detection algorithms used in our future work to design ankle-foot prostheses for the below-knee amputees to walk on different terrains.

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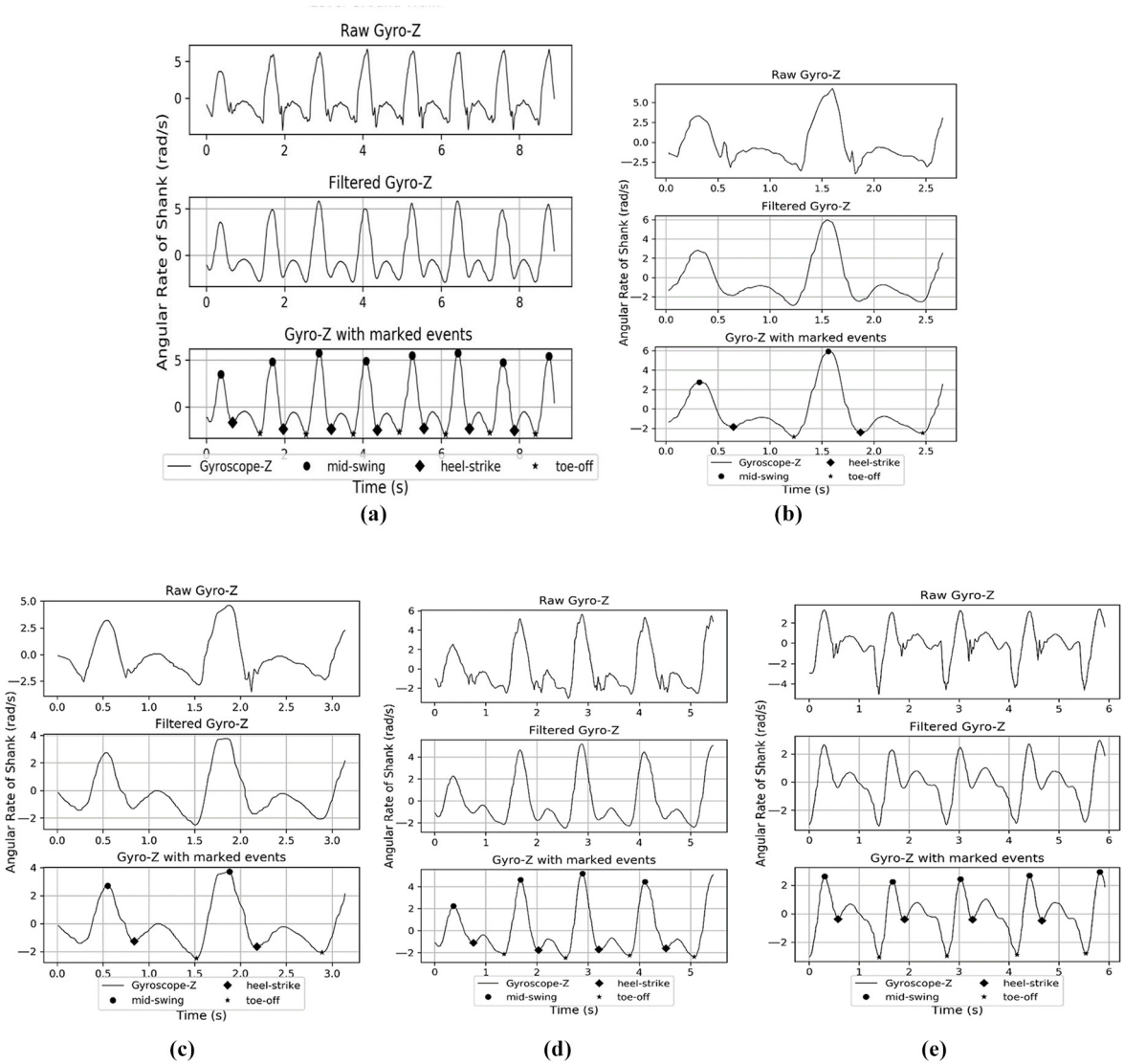
Figure 9 Gyroscope readings of IMU Sensor attached at the shank for different gait patterns



Notes: (a) Level Ground Walk; (b) Ramp Descent Walk; (c) Ramp Ascent Walk; (d) Stair Descent Walk; (e) Stair Ascent Walk

Figure 10 Real-time gait phase detection using a single IMU sensor

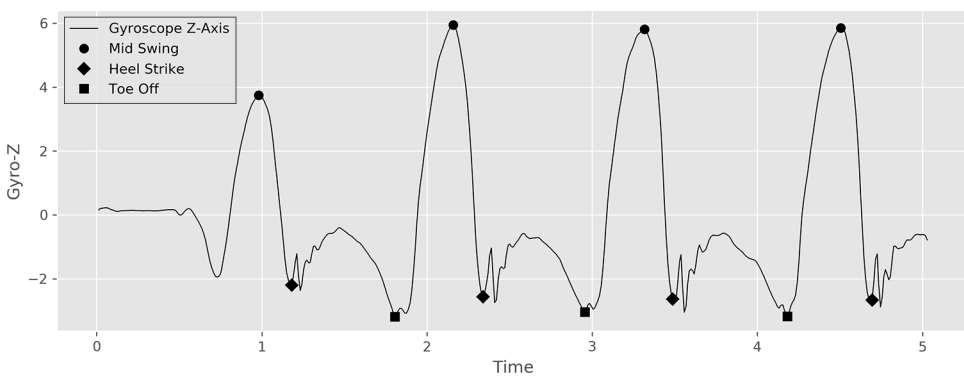


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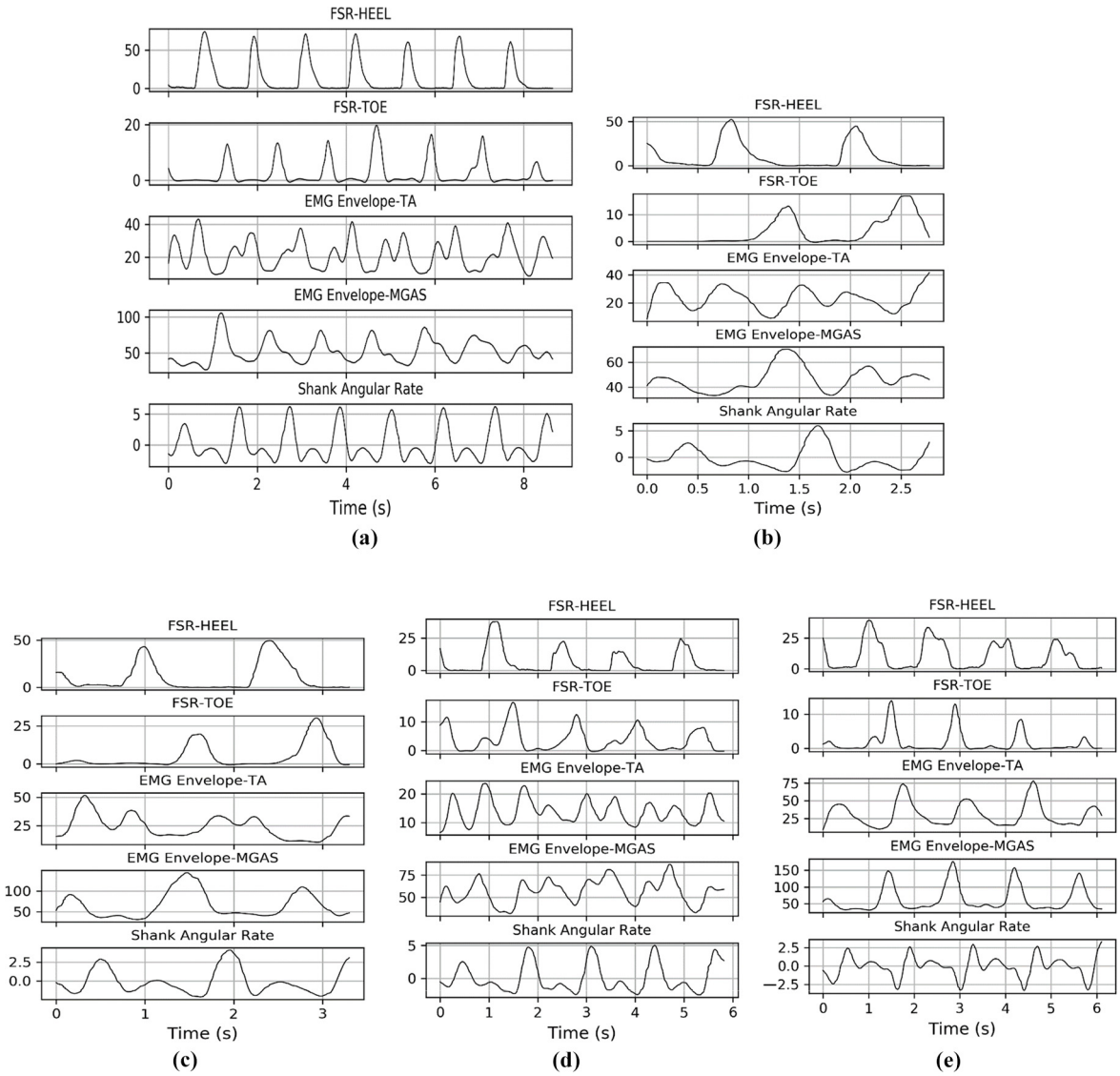
Figure 11 Synchronization of FSR, IMU and EMG sensors data for different gait patterns



Notes: (a) LG Terrain; (b) RD Terrain; (c) RA Terrain; (d) SD Terrain; (e) SA Terrain

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